



Daniel Watts

Senior Software Engineer ·
Full-Stack Web + Applied AI

CONTACT

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SKILLS

Expert

PHP (4 to 8.3) · JavaScript · SQL
(MariaDB/MySQL) · MVC
architecture · REST API
integration and design · UI/UX
and workflow design

Production experience

nginx, Caddy, Apache · SAML
2.0 SSO (Azure Entra ID) · LDAP
· NIST 800-171/CUI · Git/GitHub
· Agile/Scrum

Daily AI tools

Claude Code (custom agent
skills, subagents) · Gemini

Working knowledge, AI- accelerated

Python · Laravel · Symfony ·
PostgreSQL · RAG (Ollama,
embeddings) · MCP

EDUCATION & RECOGNITION

A.A.S. in Computer Animation, with honors

Colorado Institute of Art,
Denver · Best graduating
portfolio award

NCAR Special Recognition Award, February 2026

As Project Lead and Main
Developer of the NRIT Visit
App; nominated by the
Director of Research IT for
work "of unusual importance
to the execution of UCAR's
programs"

PROFILE

Senior engineer with 20+ years building and modernizing production web applications, and a current focus on applied AI engineering. Started in design, so the work spans the whole experience: UI/UX and front end through back end and infrastructure. At NSF NCAR, sole developer of an organization-wide visitor management platform: architected it, designed its workflows and interfaces, containerized it, shipped 60+ production releases, and deployed it to a NIST 800-171 CUI-compliant environment. Builds with AI daily: custom Claude Code agent skills, AI-assisted legacy modernization, and RAG retrieval, plus the organization's AI guidance documentation.

EXPERIENCE

Software Engineer/Programmer II · UCAR / NSF National Center for Atmospheric
Research · Boulder, CO

2019 to Present

Joined as a contract LAMP developer in late 2019; converted to staff in 2020. Sole developer of RAL-IT's administrative application suite (visits, purchasing, equipment, accounts). Project Lead and Main Developer of the NRIT Visit App, the visitor management platform used across all NSF NCAR labs, recognized with an NCAR Special Recognition Award (2026).

Platform engineering

- Architected and built RAL's administrative application platform from scratch, replacing two undocumented legacy systems and maintaining the rest of the legacy suite; in continuous production since 2021 with 60+ versioned releases shipped solo, end to end.
- Transformed RAL's single-lab tool into a multi-tenant platform serving 8 NCAR labs through a config-driven architecture: per-lab feature flags, field overrides, email templates, and admin roles. Onboarded each lab personally: requirements, demos, training, and sign-off.
- Executed the production cutover from a runbook he authored: migrated 3,113 visit records and 8,799 users across hosts with cron freezes, smoke tests, 35 preserved legacy URLs, and DNS repointing. Zero data loss.
- Modernized the stack end to end: PHP 7.1 to 8.3, Composer/Symfony adoption, SVN to Git/GitHub, and XAMPP to a containerized dev environment (Docker Compose, rootless Podman with systemd Quadlet units).
- Integrated enterprise systems via automated cron pipelines: Workday, the UCAR person directory (~10,000 accounts), Request Tracker (two instances), and enterprise auth, with a 25+ template notification email system.

Applied AI engineering

- Prototyped a self-hosted RAG Slack bot answering questions over the organization's Confluence wiki: hybrid full-text plus embedding retrieval with hierarchy-aware chunking, read-only API ingestion, incremental sync, local LLM via Ollama, and an 81-test pytest suite. Indexes 793 wiki pages in ~13 minutes on CPU.
- Authored custom Claude Code agent skills that maintain organizational documentation: each parses unstructured input (websites, GitHub READMEs, wiki and KB pages), classifies findings against tracker state, and proposes reviewed edits before writing.
- Rebuilt a 20-document developer doc set with AI-assisted workflows that ingested local notes, the full project Slack thread, and outdated wiki documentation, then verified every claim against the codebase, catching stale facts the old wiki had wrong.
- Documented a 26,000-line undocumented legacy ORM and executed a zero-behavior-change PHP 8 upgrade of it in under a day.
- Wrote NCAR's AI guidance: the "State of AI Tools" landscape document and the "Using AI at NCAR" policy companion, plus trackers of organizational AI tools and projects.

Security and identity

- Delivered SAML 2.0 SSO against Azure Entra ID (SimpleSAMLphp) in production: assisted a coworker's rollout on the first app, then delivered the second app's SSO solo, AI-assisted, from local to production in five days using a zero-downtime pattern: deploy dark, validate trust, flip one config value, instant rollback.
- Supported the Visit App's NIST 800-171 assessment for CUI/PII handling: authored the application data inventory, presented data handling to security and legal stakeholders, and deployed the app to the CUI-compliant environment built by the sysadmin team.
- Hardened the Visit App: app-wide prepared statements (~40 call sites), CSP headers, secrets extraction, CVE patching.
- Found a cross-division authorization leak while acceptance-testing the organization's new identity platform.

- Built a PHP library abstracting directory lookups to carry five applications through an enterprise identity-platform migration (PeopleDB to LDAP).

Leadership

- Mentored a new software engineer from onboarding through independent feature delivery; served on the hiring committee that recruited him.
- Helped design and introduce the team's scrum process; authored three published organization-wide working standards; facilitated recurring team ceremonies.
- Delivered strategic analyses to leadership, including a service-desk migration strategy and product positioning for the Visit App.

Senior Web Application Developer · Sounds True · Louisville, CO 2017 to 2019

Principal developer for the custom online courses and events platform; managed media delivery infrastructure; mentored junior developers.

- Led development of the course platform from prototype (2016) to production (2017): custom JavaScript single-page app, LAMP back end on AWS EC2/RDS, WordPress authoring admin, OAuth via Magento REST API. The platform handled 30,000 users in a single day for live broadcasts.
- Built a migration app that moved 100+ courses and 130,000 user accounts to the new platform, operable by front-end developers themselves.
- Engineered a schedulable multi-video delivery player (JavaScript, AWS S3) that automated a manual broadcast process, saving 40-50 hours per month.

Web Application Developer · Sounds True 2011 to 2017

Led the front-end team, later moved to back-end; Agile/Scrum throughout.

- Implemented the company-wide online media management system (JW Platform, AWS S3): roughly \$120,000 in annual savings and 30+ TB/month of adaptive streaming.
- Built and managed a live streaming broadcast system supporting 30,000+ concurrent viewers (PHP, JavaScript, Cassandra, AWS, Akamai).
- Replatformed three major systems including two ecommerce sites; conceived a radio app that generated ~\$40,000/year in new revenue; shipped 200+ pages, microsites, and web apps.

Web Developer · Sounds True 2006 to 2011

- Launched the company's first digital product (a custom online audio download store) and its first live online video broadcast.
- Built ecommerce front ends, email subscription management, analytics tracking, and digital product tools; managed email marketing to lists of up to 100,000, averaging \$40,000/month in direct revenue.
- Cut course design implementation from 2-3 days to 2-3 hours per course by introducing templated design assets and automated delivery.

Earlier: TinkerMedia (founder, freelance web development, 20+ clients), 2002 to 2006; Zeitgeist Design (multimedia design), 2000 to 2001; E-Vitro/DigitalMed (digital illustration), 1999 to 2000.